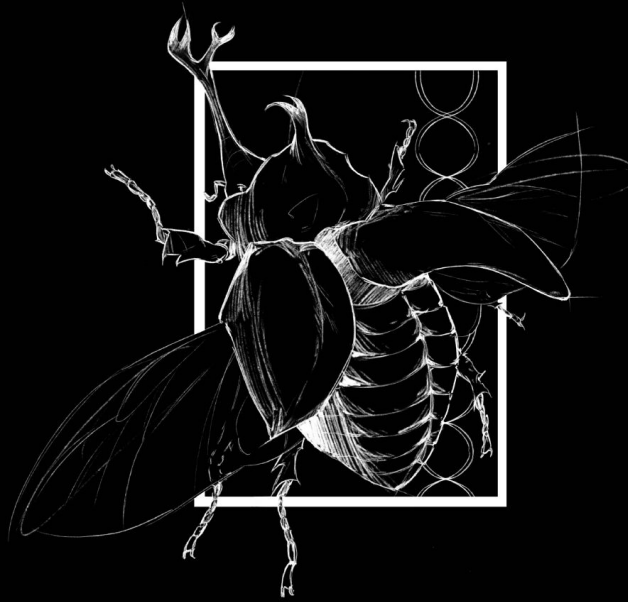




Backgrounds

Bats rely heavily on echolocation to navigate their environment and locate prey, making call duration an important functional trait. With call durations varying from species to species. With shorter calls being used in more tight regions and longer calls in more open regions. At the same time, bats display a wide range of flight morphologies, including varying wing shape, size, and membrane structure, which are often associated with how and where they fly. With bats that fly in cluttered environments tending to have shorter, more maneuverable wings, while those in open spaces having longer, more efficient wings. Because both echolocation and flight are closely tied to how bats interact with their environment, it has been suggested that these traits may be linked. However, it is still unclear whether echolocation call duration is directly influenced by flight morphology or if it is primarily driven by body size and evolutionary history. This study aims to test whether traits like forearm length, aspect ratio, and uropatagium area can predict variation in call duration across bat species, while accounting for shared ancestry using phylogenetic methods.

Study Design & AI

I investigated whether echolocation call duration in bats is better explained by flight morphology or body size. Morphological traits (forearm length, wingspan, wing area, and uropatagium area) were measured from museum specimen images, and aspect ratio was calculated. Call duration data were compiled from published sources, and species were matched to a Chiroptera phylogeny. Only species with complete data were included. I'm looking at how bat flight morphology specifically wing aspect ratio and uropatagium area has evolved alongside echolocation call traits. Using museum specimens for physical measurements, Images collected from Vertnet and GBIF

Phylogenetically controlled models (PGLS) were used to account for shared ancestry, with forearm length included to control for body size. Model comparisons using AIC were used to determine the best predictors of call duration.

AI tools were used to organize ideas, organize data, and generate initial R code.

Model Fitting

I used both standard linear models (OLS) and phylogenetically controlled models (PGLS) to test how morphology relates to echolocation call duration in bats. The reason for using PGLS is that species are not independent of each other—closely related bats may have similar traits simply because they share a common ancestor, not because those traits evolved independently. If phylogeny is ignored, it can lead to misleading results.

The PGLS models use information from the phylogenetic tree to account for these relationships by estimating Pagel's lambda, which measures how much phylogeny influences the trait. This allowed me to test whether variables like forearm length, aspect ratio, and uropatagium area still predict call duration after controlling for shared ancestry.

I ran a full model including all predictors, as well as reduced models to compare which variables best explain variation in call duration. Model fit was evaluated using AIC, and phylogenetic signal was tested separately to confirm that accounting for phylogeny was necessary. Overall, this approach provides a more accurate understanding of how morphology and body size relate to echolocation in bats.



Data & Results

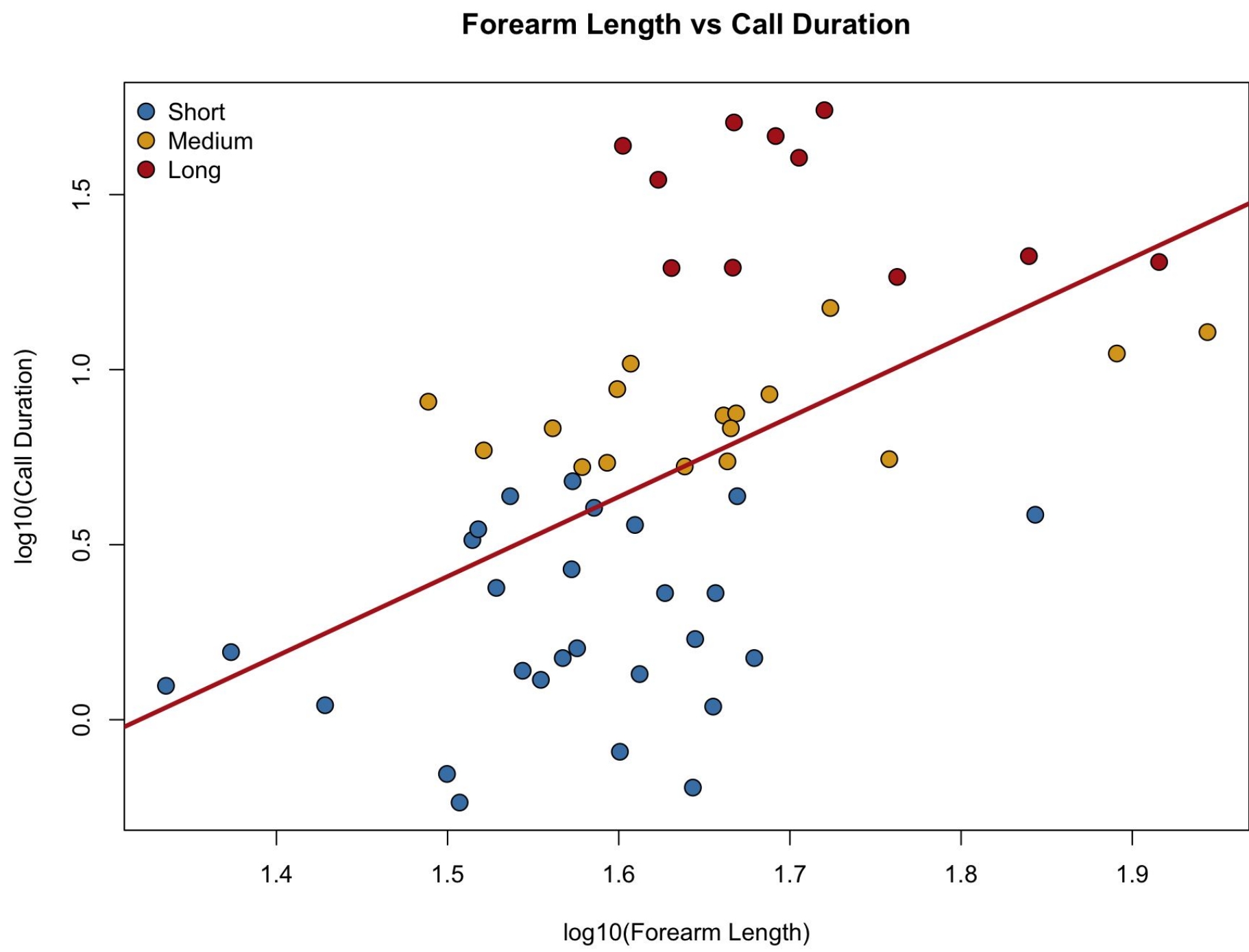


Figure 1: Relationship between forearm length and echolocation call duration in bats. Scatterplots show the association between log-transformed forearm length (mm) and log-transformed call duration (ms) across sampled bat species. Each point represents a single species included in the phylogenetic analysis. Forearm length serves as a proxy for overall body size. The positive trend indicates that larger-bodied bats tend to produce longer-duration echolocation calls. This pattern is consistent with results from both non-phylogenetic (OLS) and phylogenetically controlled (PGLS) models, where forearm length was identified as the strongest predictor of call duration.

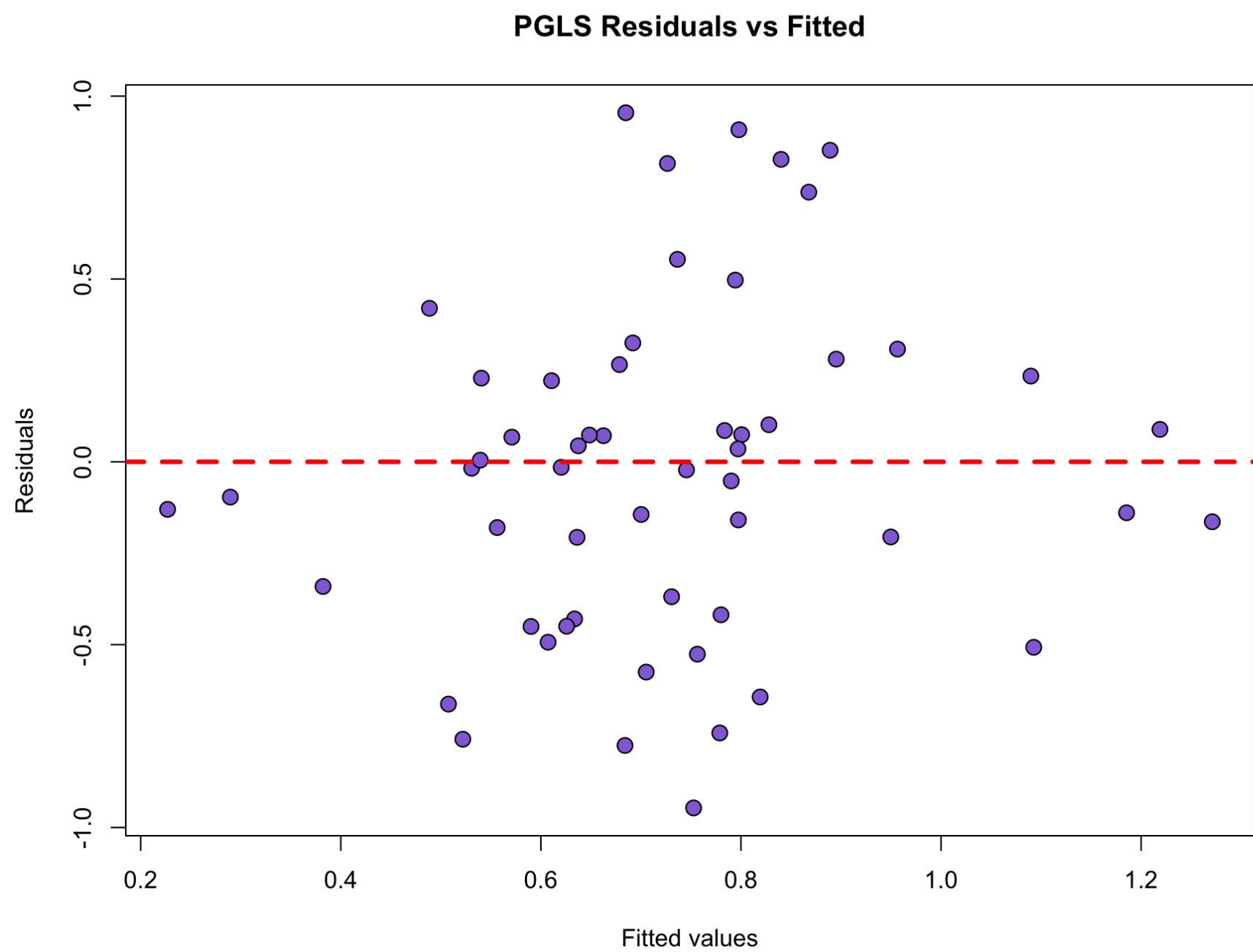


Figure 3. Diagnostic plot of PGLS residuals versus fitted values. Residuals from the phylogenetic generalized least squares (PGLS) model are plotted against fitted values to assess model assumptions. The absence of a clear pattern and the roughly symmetric distribution of residuals around zero suggest that assumptions of linearity and homoscedasticity are reasonably met.

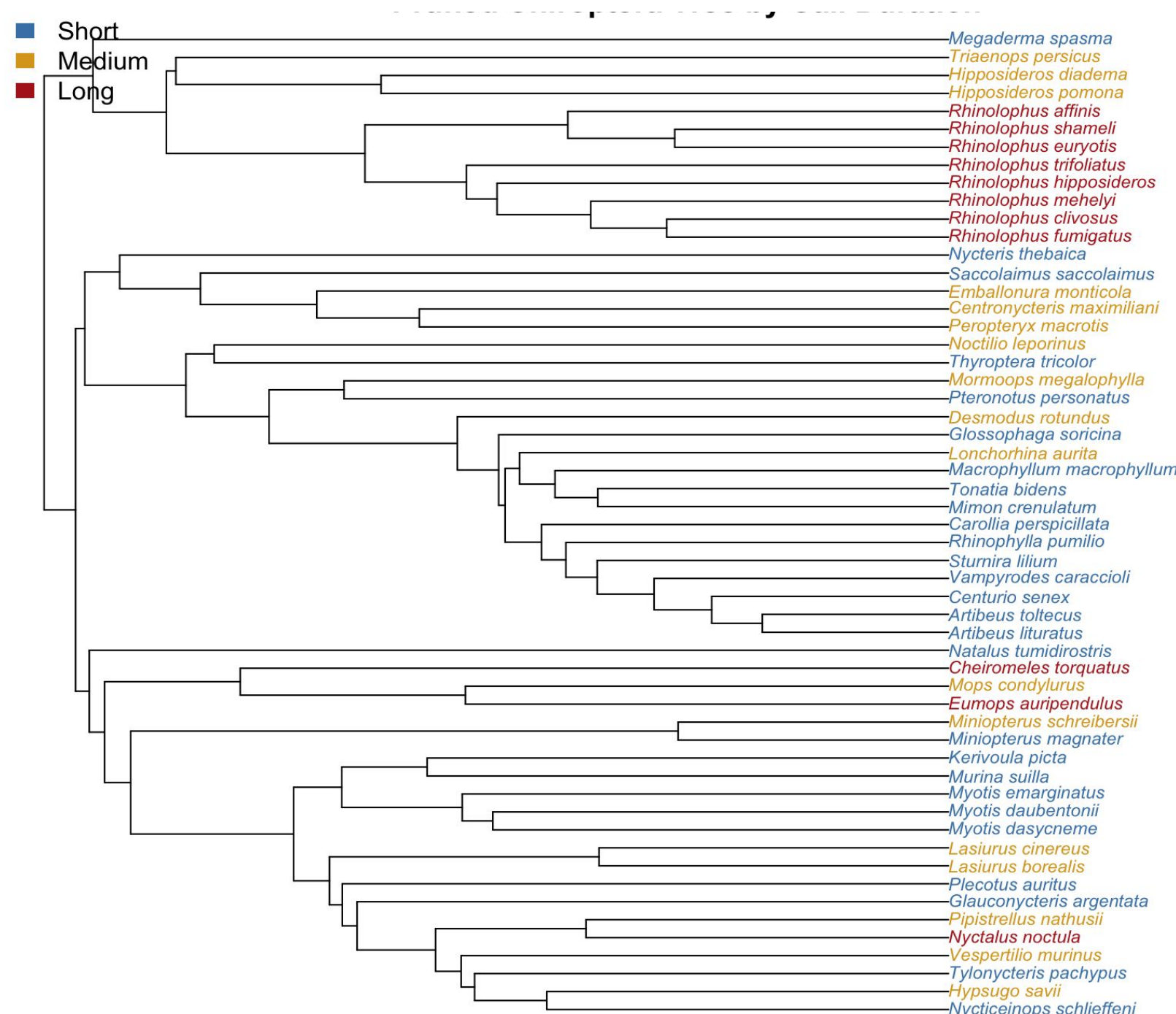


Figure 2. Phylogenetic relationships of sampled bat species with call duration mapped onto the tips. The phylogenetic tree represents evolutionary relationships among bat species included in the analysis. Tip colors correspond to categorized echolocation call durations (short, medium, long), illustrating how acoustic traits are distributed across the phylogeny. The clustering of similar call durations among closely related species highlights strong phylogenetic signal, supporting the use of phylogenetically controlled models.

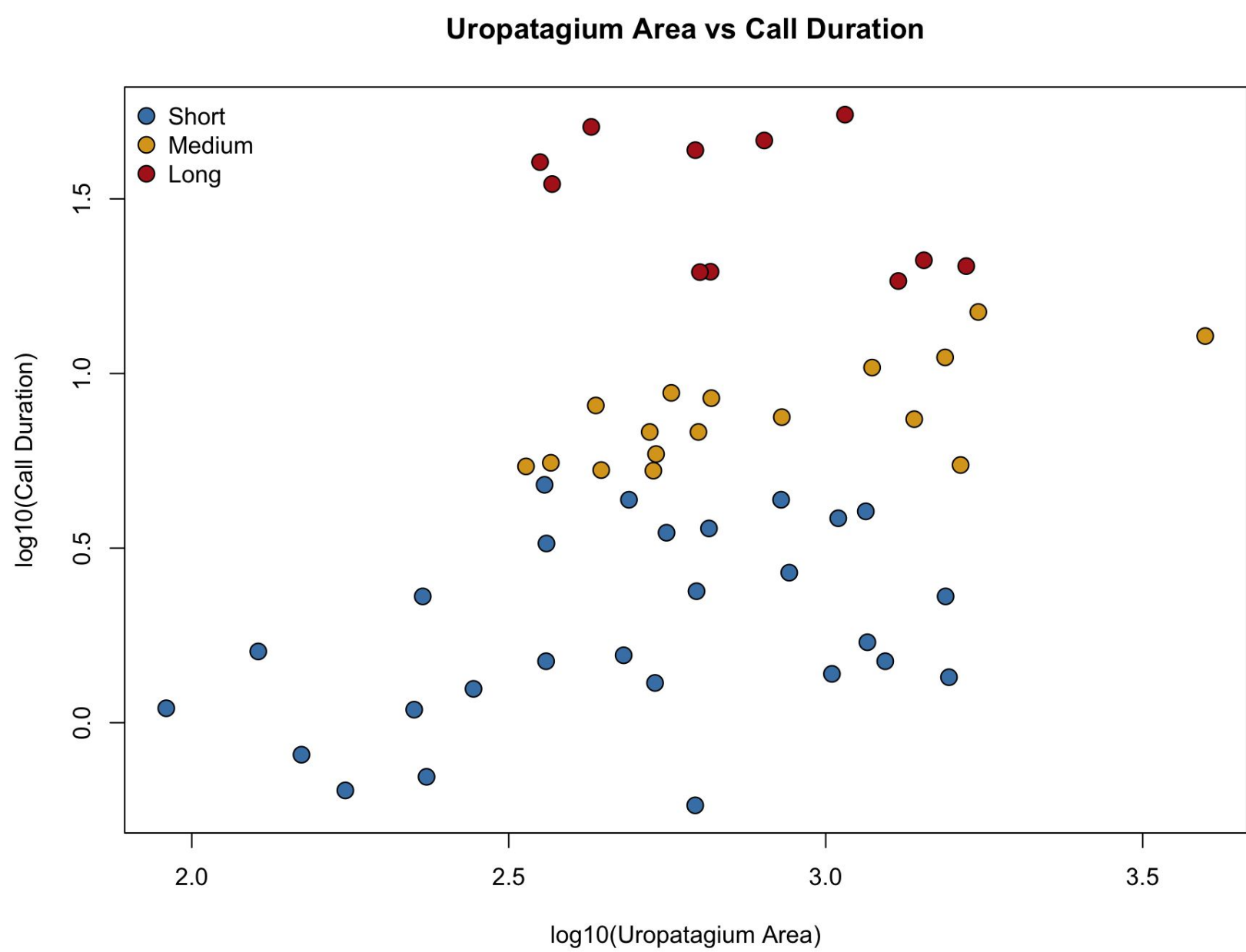


Figure 4. This figure shows the relationship between uropatagium area and echolocation call duration across bat species. Each point represents a single species, with both variables shown on a log scale. Overall, there is no clear pattern or strong trend in the data, as the points are fairly scattered. This suggests that uropatagium area is not a strong predictor of call duration, which is consistent with the statistical results showing no significant relationship between these variables.

Discussions

My results suggest that echolocation call duration in bats is driven more by body size than by flight morphology. With Forearm length (being used as a proxy for size) being the most consistent predictor of call duration in both non-phylogenetic and phylogenetically controlled models. This shows that larger bats generally produce longer calls, which fits with basic allometric scaling expectations.

In contrast, I didn't find a significant relationship between call duration and either aspect ratio or uropatagium area. Even though these traits are tied to flight performance, they don't seem to explain variation in call duration once phylogeny is taken into account. This suggests that the link between flight morphology and echolocation duration may not be as strong as expected, at least for the traits I measured.

One possible explanation is that call duration is mainly limited by body size, like how bats produce sound or how much energy they can use, rather than their wing shape. It's also possible that wing structure affects other parts of echolocation, like pitch or call pattern, instead of how long the call lasts. On top of that, factors like habitat or how bats hunt could also influence call duration, but those weren't included in this study.

Closely related species tend to have similar call duration, as shown by a strong phylogenetic single. Highlighting how its important to account for shared ancestry when analyzing traits.

There are a few limitations to consider. The measurements were taken from images, so they're somewhat rough and could introduce error, especially for traits like uropatagium area. The sample size was also relatively small (~60 species), which may reduce the ability to detect weaker relationships.

Overall, these results suggest that body size and evolutionary history play a bigger role in shaping call duration than flight morphology. More data and more precise measurements could help clarify whether smaller effects exist.

Conclusions & Acknowledgements

Conclusion: Call duration in bats is best explained by body size rather than flight morphology. Forearm length was the strongest predictor, while aspect ratio and uropatagium area were not significant after accounting for phylogeny. Overall, this suggests that allometry and evolutionary history play a larger role in shaping call duration than wing structure.